THE IMPACT OF GENERATIVE AI ON CRITICAL THINKING, PROBLEM-SOLVING SKILLS AND PERSONAL USE AMONG UNDERGRADUATE STUDENTS

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INTRODUCTION

The use of generative artificial intelligence (AI) has increased significantly in recent years, particularly in education. Tools like ChatGPT, Google Gemini, and Claude are widely used by students for completing assignments, solving problems, and understanding concepts, making academic tasks faster and easier. However, this growing reliance on AI raises concerns about its impact on essential skills such as critical thinking and problem-solving. While AI can support learning by simplifying complex topics and improving efficiency, excessive dependence may reduce students' ability to think independently and solve problems on their own. Therefore, this study aims to examine whether AI acts as a supportive learning tool or leads to increased dependency, affecting the cognitive development of undergraduate students.

Literature review

Study 1: Role of Usage Behaviour and Self-Regulation

Study 2: Impact on Critical Thinking Skills

Study 3: AI and Problem-Solving Ability

STUDY 1: Role of Usage Behaviour and Self-Regulation

The study by Alghamdi (2025) examines how generative AI influences higher-order cognitive skills among university students. It highlights that AI can act as a “cognitive scaffold” supporting learning, but may also lead to cognitive dependency if overused. The findings show that STEM students have a more positive perception of AI in enhancing critical thinking compared to non-STEM students, while all students moderately agree that AI supports creativity. A key insight is that students with higher AI self-efficacy perceive AI as a tool for learning rather than a shortcut. The study concludes that AI can enhance cognitive development only when students actively engage with it and are trained to critically evaluate its outputs.

STUDY 2: Impact on Critical Thinking Skills

The study by Brazão and Tinoca (2025) analyzes how AI chatbots affect students' critical thinking through different levels of questioning. It finds that AI can support higher-order thinking when students actively engage by asking applied and critical questions. However, over-reliance on AI leads to passive learning and reduced independent thinking. The study emphasizes that the impact of AI is highly dependent on how students use it. Proper guidance and structured learning frameworks are necessary to ensure that AI enhances rather than weakens critical thinking abilities.

STUDY 3: AI and Problem-Solving Ability

The study by dos Santos explores how AI chatbots like ChatGPT can support problem-solving in complex subjects such as Chemistry. It shows that AI can act as an interactive tutor by providing personalized explanations and encouraging students to think through questions. The study also finds that the effectiveness of AI depends on how students interact with it, especially their ability to ask meaningful prompts. However, AI may still produce incorrect responses, making critical thinking essential. The study concludes that AI has strong potential to improve problem-solving skills if used actively and carefully.

Research gap

- Most existing studies focus on students' perceptions of AI, rather than measuring its actual impact on their academic performance and cognitive skills
- There is a lack of research that examines critical thinking and problem-solving together, even though both are closely interconnected skills in education
- Current research is largely limited to specific disciplines such as STEM or Chemistry, and does not represent the broader undergraduate student population
- AI usage is often treated as a simple yes/no variable, without considering important factors like frequency of use or level of dependency
- There is a shortage of small-scale pilot studies that provide detailed insights into student behavior at the undergraduate level
- Existing studies offer limited practical and data-driven guidance, making it unclear how students can effectively use AI without becoming overly dependent

Objectives

1. To understand how the use of generative AI affects the way undergraduate students think critically.
2. To explore whether AI tools help students improve their problem-solving skills or make them rely too much on ready-made answers.
3. To examine if students who use AI more frequently tend to become dependent on it for completing their academic work.
4. To find out whether AI acts as a helpful learning support or reduces students' ability to think independently.
5. To compare how students with different levels of AI usage differ in their thinking and problem-solving abilities.
6. To study how different ways of using AI influence the balance between learning independently and taking help from technology.

Methodology

Hypothesis

Critical Thinking

- H_0: No significant relationship with AI usage
- H_1: Significant relationship exists

Problem Solving

- H_0: No significant impact of AI
- H_1: Significant impact exists

Personal Use

- H_0: No significant impact on personal life
- H_1: Significant impact exists

Data Type

- Primary data (first-hand)
- Collected through online questionnaire (Google Forms)

Sampling Design

- Population: Undergraduate students
- Sample Size: 50 respondents
- Method:Convenience sampling

Data Collection Method

- Structured questionnaire (Likert scale 1–5)
- Online survey via digital platforms
- Advantages: Quick, cost-effective, uniform responses

Data presentation

In the following attached excel sheet the data is organised in a clear and structured format in table form for frequency distribution along with visual representation using graphs and pie charts.

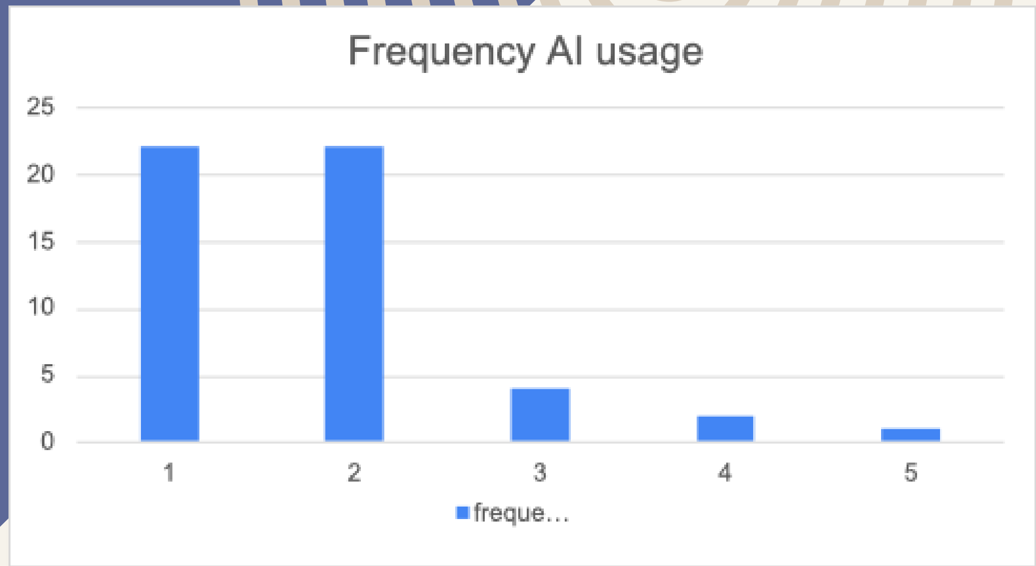
https://excel.cloud.microsoft/open/onedrive/?docId=7F88672BD1ED4F02%21s930b70ebab654dd9829e50980c32d54d&driveId=7F88672BD1ED4F02

Frequency	Strongly agree	Agree	Neutral	Disagree	Strongly disagree
Frequency(AI Usage)	22	22	4	2	1
Frequency(Problem Solving)	5	22	16	8	0
Critical thinking	7	21	15	6	2
Personal use	12	16	10	11	2

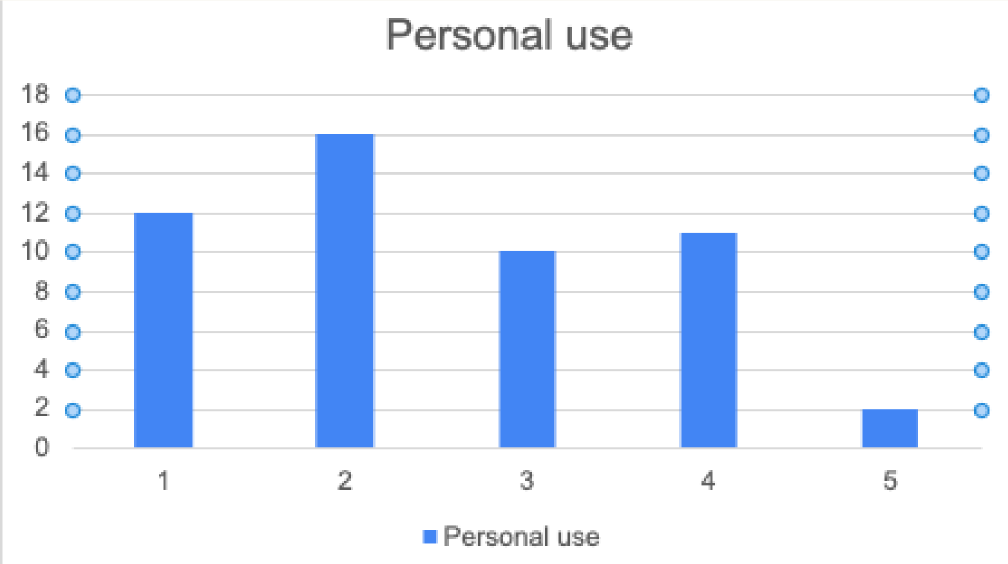
The above given data shows the frequency distribution of the sample collected from a sample of 50 undergraduate students through the google form

FORM:

https://docs.google.com/forms/d/e/1FAIpQLSd9nhV5ukE7V3kzyhj_-81ojyxOmotJ0zL2r9Uhiw4NJWZYFg/viewform?usp=dialog



Majority of respondents frequently use AI tools



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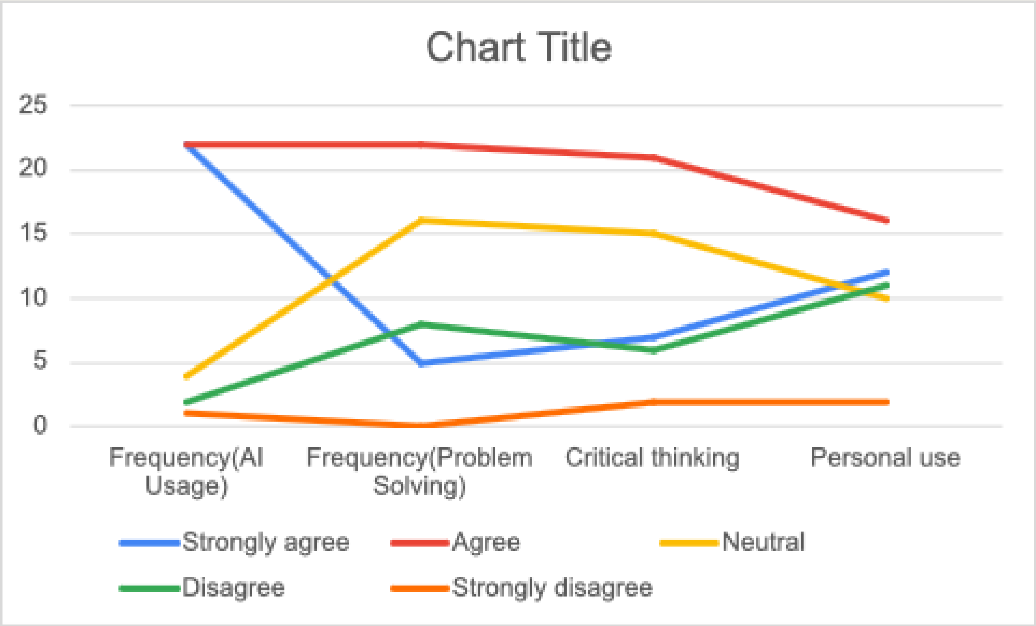
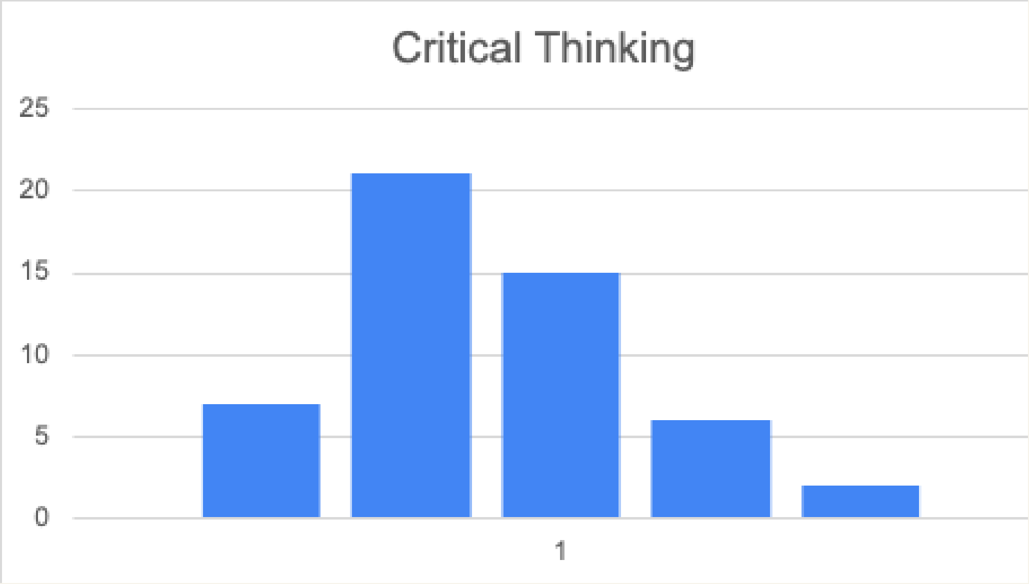
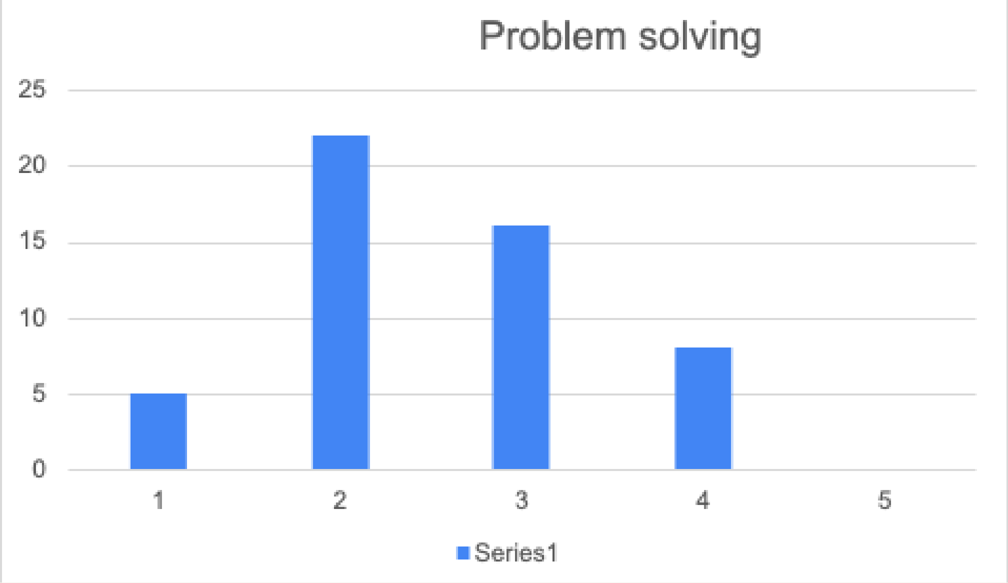


AI impact is strongest on personal use, weakest on problem solving skills



Responses are mixed, indicating no strong effect in critical thinking



AI shows limited impact on problem-solving ability

Descriptive Analysis

AI Elements	Average	median	mode	standard deviation	variance
AI usage	3.965686275	4	4	0.782655109	0.61254902
Problem solving	3.544117647	3.5	3.5	0.55835894	0.311764706
Critical thinking	3.289215686	3.25	3	0.691687337	0.478431373
Personal use	3.593137255	4	4	1.004938785	1.009901961

Interpretation of Descriptive Statistics on the basis of 50 respondents:

- The mean values between (3–4) indicate that most respondents agree that they use AI frequently
- Median and mode being close to 4 shows majority responses are “Agree” or “Strongly Agree”
- Personal use has highest mean, indicating strongest usage area
- Standard deviation is relatively low (0.5–1), meaning:
- Responses are not highly spread out as most students have similar opinions
- In simple terms:
- Students consistently report moderate to high usage of AI, especially for personal purposes.

Statistical Techniques

Correlation Analysis

- Measures the strength and direction of relationship between two variables
- Value of correlation (r) ranges from -1 to +1

  +1 → Strong positive relationship

  -1 → Strong negative relationshi

  0 → No relationship
- Used to analyze relationship between AI usage and other factors

$$r = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2 \cdot \sum(Y_i - \bar{Y})^2}}$$

t-test (Hypothesis Testing)

- Used to check whether results are statistically significant
- Helps in deciding whether to:
- Accept or reject the hypothesis
- Based on:
- p-value (< 0.05 = significant)
- Ensures that results are not due to chance

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Results and Interpretation

Variable	R^2	p-value	Conclusion
Personal Use	0.327	<0.05	Significant
Critical Thinking	0.013	>0.05	Not significant
Problem Solving	0.026	>0.05	Not significant

Correlation analysis

AI & Personal Use → r ≈ 0.51
Moderate positive relationship
AI & Problem Solving → r ≈ 0.18
Weak positive relationship
AI & Critical Thinking → r ≈ -0.11
Weak negative relationship

- AI usage has a strongest relationship with personal use
- Very weak relationship with problem-solving ability
- Slight negative relationship with critical thinking

t-test (Hypothesis Testing)

Decision Rule

- $p < 0.05 \rightarrow$ Reject H_0
- $p > 0.05 \rightarrow$ Accept H_0

Final Decisions

- **Personal Use** → Reject H_0
- (Significant impact of AI)
- **Critical Thinking** → Accept H_0
- (No significant impact)
- **Problem Solving** → Accept H_0
- (No significant impact)

- AI is widely used and accepted by students
- It has a clear impact on personal use
- However, it does not significantly improve higher-level skills

Final Insight:

- AI is being used more for convenience rather than for developing critical thinking or problem-solving abilities.

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Findings

- Majority of students from the 50 respondents frequently use AI tools in daily activities
- Descriptive statistics indicate high agreement on AI usage
- AI shows a moderate positive impact on personal use
- Regression results confirm significant impact on personal use
- AI has a weak and insignificant relationship with:
  - Critical thinking
  - Problem-solving ability
- Slight negative relationship with critical thinking, indicating reduced independent thinking
- Hypothesis results:
  - Rejected for personal use
  - Accepted for critical thinking & problem-solving
- Responses are consistent

Overall: AI is mainly used for convenience rather than skill development

Limitations of the Study

Data Limitations

- Based on self-reported primary data (may include bias)
- Responses may reflect perception, not actual behavior
- Limited variables considered (excluded performance, creativity, etc.)

Sample Size

- Small sample size (pilot study)
- May not represent all students or backgrounds
- Limits generalization of results

Time Constraints

- Study conducted within a limited time period
- Restricted depth of analysis
- Advanced techniques and larger data collection not possible

Scope for Future Research

Increase in Sample Size
- Include a larger and more diverse sample
- Cover different colleges, courses, and age groups

Inclusion of Additional Variables
- Academic performance
- Creativity
- Decision-making ability
- Learning outcomes

Advanced Statistical Techniques
- Multiple regression
- Factor analysis
- ANOVA
- For deeper and more accurate analysis

Comparative Studies
- Compare frequent vs non-AI users
- Across different institutions or regions

Practical Application-Based Research
- Focus on real academic tasks instead of only surveys
- Provides more realistic and reliable results

Conclusion

The study aimed to examine the impact of AI usage on students, focusing on personal use, critical thinking, and problem-solving ability. The analysis revealed that AI has a significant impact on personal use, as students frequently rely on it for convenience and efficiency in daily tasks. However, it showed no significant impact on critical thinking and problem-solving skills, indicating that AI does not necessarily enhance higher-order cognitive abilities. Overall, the study concludes that while AI is widely used and beneficial for saving time, it is primarily a convenience tool rather than a skill-development tool. Therefore, AI should be used in a balanced manner, supporting learning without replacing independent thinking and problem-solving.

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THANK YOU